



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CULTIVATING OPERATIONS

Page 1 of 5

SOP No:	3-CUL-001	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

TO BE CUSTOMISED

LIST OF ABBREVIATIONS

GAP	Good Agriculture Practice
GSP	Good Storage Practice
AR	Authorised Representative
SOP	Standard Operating Procedure

1. OBJECTIVES:

To develop and implement a facility with trained personnel to ensure all cultivation operations will meet requirements and standards of SAHPRA, and that no operation will be performed without, adherence to guidelines as set out by Good Agricultural Practices.

2. SCOPE:

This SOP explains the operations of medicinal cannabis cultivation procedures.

3. RESPONSIBILITY:

AR	Authorised Representative
HG	Head Grower

4. POLICY:

Good agricultural practices for the cultivation of medicinal cannabis must be applied, as determined by the Department of Agriculture, Forestry and Fisheries (DAFF) for plants intended for human consumption, and The Medicines and Related Substances Act 101 of 1965 to ensure all products produced meet the required standards.

5. PROCEDURE:

5.1 The site will either use an indoor or outdoor growing facility.

5.2 The site will select the type of cultivation method:

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CULTIVATING OPERATIONS

Page 2 of 5

SOP No:	3-CUL-001	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

- Hydroponics
- Aquaponics
- Aeroponics

5.3 The site will list the nutrients and growing medium to be utilised.

5.4 The method of preparation of the growing medium will be specified by the site.

5.5 Tools used in the facility may include:

- Growing pots
- Stainless steel racks
- Pruning scissors
- PH meter
- Mixing jars
- Conductivity meter
- Jiffy pellets
- Measuring cylinders

5.6 All tools will be cleaned and sanitised prior to use.

5.7 The PH and conductivity meters will be calibrated prior to use and externally recalibrated on an annual basis.

5.8 Batch coded seeds are planted into the specified growing medium and grown for between 8 - 14 days to reach the germination stage.

5.9 Purified water is used throughout the process.

5.10 The seedlings are transferred to bigger pots for initiation of the vegetative growth stage.

5.11 The daily dosing of nutrients per plant is recorded.

5.12 The plants will grow for between 3 to 6 weeks, dependant on the strain and operating requirements.

5.13 The HG will determine when each plant is ready for harvesting.

5.14 The SOP: 3-CUL-002 Procedure for Sexing a Cannabis Plant. Will be followed in order to ensure only female plants are accepted for further processing.

6. Attachments

Attachment 6.1 SOP Amendment History

Attachment 6.2 Distribution List of SOPs

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



Authorised
Copy

ORIGINAL
DOCUMENT
T

PROCEDURE FOR CULTIVATING OPERATIONS

Page 3 of 5

SOP No:	3-CUL-001	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

[illegible]

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ORIGINAL
DOCUMENT
T

Page 4 of 5

SOP No:	3-CUL-001	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

DISTRIBUTION LIST OF STANDARD OPERATING PROCEDURES

[illegible]

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CULTIVATING OPERATIONS

Page 5 of 5

SOP No:	3-CUL-001	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR SEXING A CANNABIS PLANT

Page 1 of 5

SOP No:	3-CUL-002	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

LIST OF ABBREVIATIONS

GAP	Good Agriculture Practice
GSP	Good Storage Practice
AR	Authorised Representative
SOP	Standard Operating Procedure

1. OBJECTIVES:

To ensure sexing plant operations will meet the requirements and standards to select only female plants for harvesting.

2. SCOPE:

This SOP explains the operations of how to identify the gender of each plant.

3. RESPONSIBILITY:

AR	Authorised Representative
HG	Head Grower

4. POLICY:

The facility will apply Good agricultural practices for the sexing plants of medicinal cannabis, as required by the Department of Agriculture, Forestry and Fisheries (DAFF) for plants intended for human consumption, and The Medicines and Related substances Act 101 of 1965.

5. PROCEDURE:

5.1 Identification of a Male Medicinal Cannabis plant:

- 5.1.1 Male medicinal cannabis plants will grow distinct ball-like sacs, Hermaphrodite plants will also grow ball-like sacs.

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter

**STANDARD OPERATING PROCEDURE****PROCEDURE FOR SEXING A CANNABIS PLANT**

Page 2 of 5

SOP No:	3-CUL-002	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

- 5.1.2 Once they open, a fine powder is released (pollen) that is dispersed throughout the room on the slightest breeze.
- 5.1.3 The AR or HG will take care to eliminate a contamination risk when removing male plants. A cover may be used to prevent pollen from being used.
- 5.1.4 Male plants will be destroyed in a specified manner, such as placing the plant in a plastic bag and securing the bag.
- 5.1.5 Each bag will be labelled with a reject sticker and removed to the destruction area.
- 5.2 Each batch of flowering plant is grown in a separate flower room to prevent cross-contamination from male to female plants of different batches.
- 5.3 Male plants that are destroyed are recorded in the medicinal cannabis register with their batch number and weight for traceability purposes.

6. Attachments**Attachment 6.1** SOP Amendment History**Attachment 6.2** Distribution List of SOPs

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE			
PROCEDURE FOR SEXING A CANNABIS PLANT			Page 3 of 5
SOP No:	3-CUL-002	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

Attachment 6.1

SOP AMENDMENT HISTORY			
VERSION NO:	SUPERSEDES:	EFFECTIVE DATE:	REASON FOR CHANGE:
1	N/A	2018-10-01	First Version
2	1	2020-06-01	Updated for clarity

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter

STANDARD OPERATING PROCEDURE

PROCEDURE FOR SEXING A CANNABIS PLANT

Page 4 of 5

SOP No:	3-CUL-002
---------	-----------

Revision No:	2
--------------	---

Effective Date:	2020-06-01
-----------------	------------

Revise By:	2022-06-01
------------	------------

Supersedes:	1
--------------------	----------

Attachment 6.2

DISTRIBUTION LIST OF STANDARD OPERATING PROCEDURES

[illegible]

COPY NO:

Compiler:

CCB – Nico Kriek

Authorised by:

Coenraad Venter

Approved by:

Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**
T

STANDARD OPERATING PROCEDURE

PROCEDURE FOR SEXING A CANNABIS PLANT

Page 5 of 5

SOP No:	3-CUL-002	Effective Date:	2020-06-01
Revision No:	2	Supersedes:	1
Revise By:	2022-06-01		

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

Page 1 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

LIST OF ABBREVIATIONS

GAP	Good Agriculture Practice
GSP	Good Storage Practice
AR	Authorised Representative
SOP	Standard Operating Procedure

1. OBJECTIVE:

To develop and implement a facility with trained personnel to ensure all cloning operations will meet requirements and standards, and that no operation will be performed without adherence to guidelines as set out by Good Agricultural Practices.

2. SCOPE:

This SOP explains the operations of cloning procedures for a medicinal cannabis plant.

3. RESPONSIBILITY:

AR	Authorised Representative
FM	Facility Manager
HG	Head Grower

4. POLICY:

ILCO Farming will apply good agricultural practices for the cloning of medicinal cannabis, as determined by the Department of Agriculture, Forestry and Fisheries (DAFF) for plants intended for human consumption, and the Medicines and Related Substances Act 101 of 1965 to ensure all processes involved with the cloning process are adhered to.

5. PROCEDURE:

5.1 Ensure that you have a supply of fibre bags that contain coco peat or equivalent.

5.2 Select the mother plant to be cloned.

5.3 Identify the two side-shoots on each mother plant that will be cut off.

5.4 The side-shoots are cut just after the side branches node.

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter

**STANDARD OPERATING PROCEDURE****PROCEDURE FOR CLONING OPERATIONS**

Page 2 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

5.5 Take sterilised pruning scissors, measure the node to be cut, which should be at least 15cm long, and make a 45-degree angled cut.

5.6 Slit the bottom of the angled cut at least 1cm deep.

5.7 Place the cutting into purified water for 10 seconds or as required.

5.8 Take cutting out of water and place in hormone powder or hormone gel.

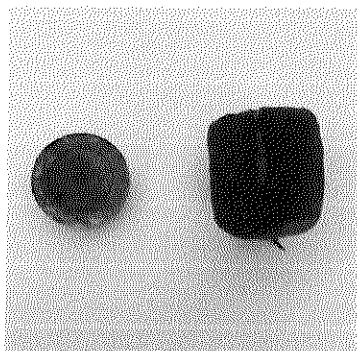
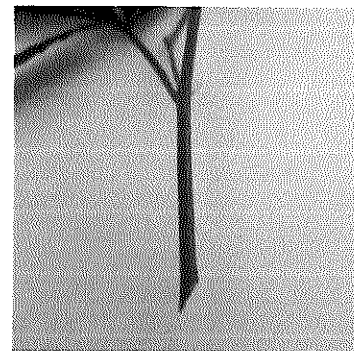
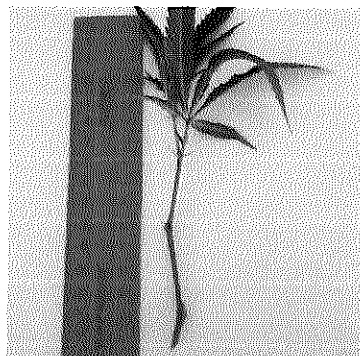
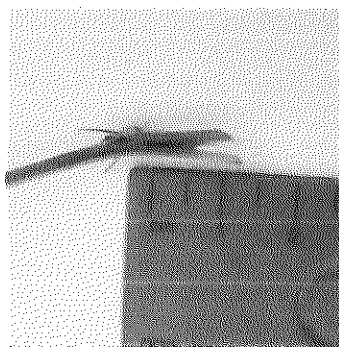
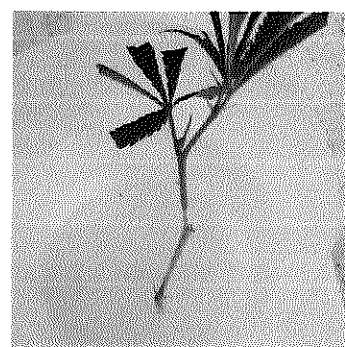
5.9 Compress the hormone powder so the slit is completely covered.

5.10 Place the cutting with the hormone powder into the coco-peat fibre bag.

5.11 Place the clone into the humidity dome.

5.12 After about 10-12 days a light feed can be beneficial including a rooting stimulant to increase root growth.

5.13 The site will transplant the clone according to SOP: 3-CUL-009 Procedure for Transplanting Operations.

**5.1. Coco-peat Bags****5.2. Mother plant****5.5. 45° Angle****5.5. 15cm cutting****5.6. 1cm slit****5.7. Place cutting in water**

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**
T

STANDARD OPERATING PROCEDURE

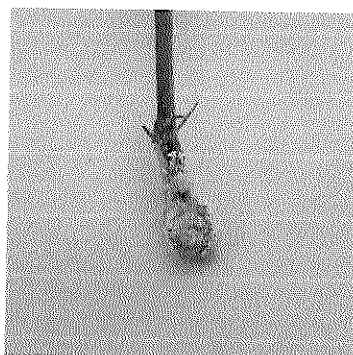
PROCEDURE FOR CLONING OPERATIONS

Page 3 of 6

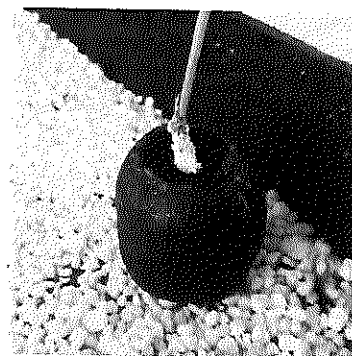
SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		



5.8. Cutting placed in
Hormone powder



5.9. Slit covered in hormone
powder



5.10. Cutting placed in fibre
bag

6. Attachments

Attachment 6.1 SOP Amendment History

Attachment 6.2 Distribution List of SOPs

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

Page 4 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

Attachment 6.1

[illegible]

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

Page 5 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

Attachment 6.2

DISTRIBUTION LIST OF STANDARD OPERATING PROCEDURES

[illegible]

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

Page 6 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**
T

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

Page 1 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

LIST OF ABBREVIATIONS

GAP	Good Agriculture Practice
GSP	Good Storage Practice
AR	Authorised Representative
SOP	Standard Operating Procedure

1. OBJECTIVE:

To develop and implement a facility with trained personnel to ensure all cloning operations will meet requirements and standards, and that no operation will be performed without adherence to guidelines as set out by Good Agricultural Practices.

2. SCOPE:

This SOP explains the operations of cloning procedures for a medicinal cannabis plant.

3. RESPONSIBILITY:

AR	Authorised Representative
FM	Facility Manager
HG	Head Grower

4. POLICY:

ILCO Farming will apply good agricultural practices for the cloning of medicinal cannabis, as determined by the Department of Agriculture, Forestry and Fisheries (DAFF) for plants intended for human consumption, and the Medicines and Related Substances Act 101 of 1965 to ensure all processes involved with the cloning process are adhered to.

5. PROCEDURE:

5.1 Ensure that you have a supply of fibre bags that contain coco peat or equivalent.

5.2 Select the mother plant to be cloned.

5.3 Identify the two side-shoots on each mother plant that will be cut off.

5.4 The side-shoots are cut just after the side branches node.

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



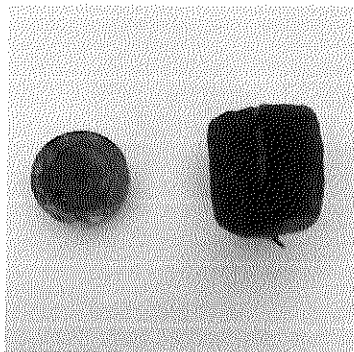
STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

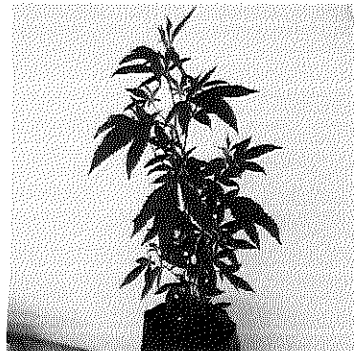
Page 2 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

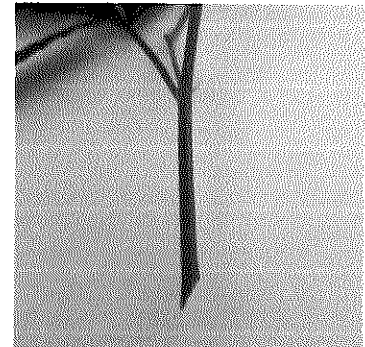
- 5.5** Take sterilised pruning scissors, measure the node to be cut, which should be at least 15cm long, and make a 45-degree angled cut.
- 5.6** Slit the bottom of the angled cut at least 1cm deep.
- 5.7** Place the cutting into purified water for 10 seconds or as required.
- 5.8** Take cutting out of water and place in hormone powder or hormone gel.
- 5.9** Compress the hormone powder so the slit is completely covered.
- 5.10** Place the cutting with the hormone powder into the coco-peat fibre bag.
- 5.11** Place the clone into the humidity dome.
- 5.12** After about 10-12 days a light feed can be beneficial including a rooting stimulant to increase root growth.
- 5.13** The site will transplant the clone according to SOP: 3-CUL-009 Procedure for Transplanting Operations.



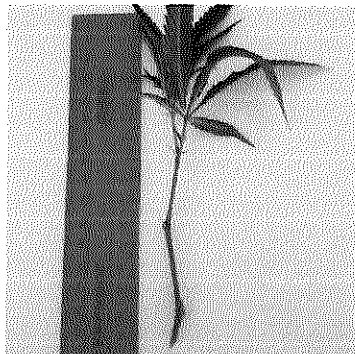
5.1. Coco-peat Bags



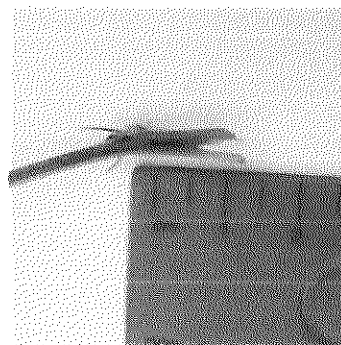
5.2. Mother plant



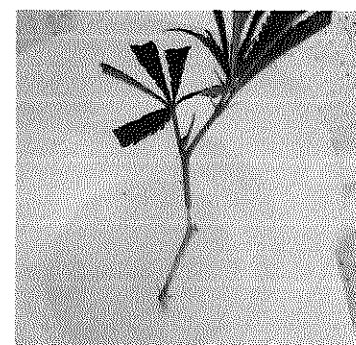
5.5. 45° Angle



5.5. 15cm cutting



5.6. 1cm slit



**5.7. Place cutting in
water**

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**
T

STANDARD OPERATING PROCEDURE

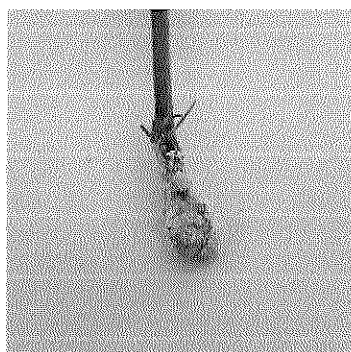
PROCEDURE FOR CLONING OPERATIONS

Page 3 of 6

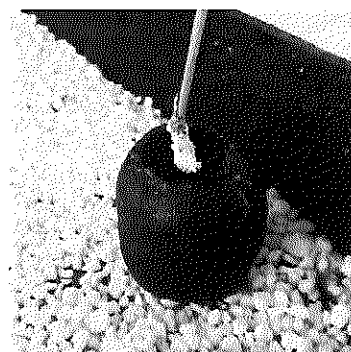
SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		



5.8. Cutting placed in
Hormone powder



5.9. Slit covered in hormone
powder



5.10. Cutting placed in fibre
bag

6. Attachments

Attachment 6.1 SOP Amendment History

Attachment 6.2 Distribution List of SOPs

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**
T

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

Page 4 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

Attachment 6.1

SOP AMENDMENT HISTORY			
VERSION NO:	SUPERSEDES:	EFFECTIVE DATE:	REASON FOR CHANGE:
1	N/A	2018-10-01	First Version
2	1	2020-06-11	Updated for clarification of process

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



Authorised
Copy

ORIGINAL
DOCUMENT
T

PROCEDURE FOR CLONING OPERATIONS

Page 5 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

Attachment 6.2

DISTRIBUTION LIST OF STANDARD OPERATING PROCEDURES

[illegible]

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT
T**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR CLONING OPERATIONS

Page 6 of 6

SOP No:	3-CUL-005	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**
T

STANDARD OPERATING PROCEDURE

PROCEDURE FOR DRYING OPERATIONS

Page 1 of 3

SOP No:	3-CUL-006	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

LIST OF ABBREVIATIONS

GAP	Good Agriculture Practice
GSP	Good Storage Practice
AR	Authorised Representative
SOP	Standard Operating Procedure

1. OBJECTIVE:

To develop and implement a facility with trained personnel to ensure all drying operations will meet requirements and standards, and that no operation will be performed without adherence to guidelines as set out by Good Agricultural Practices.

2. SCOPE:

This SOP explains the operations for drying medicinal cannabis procedures.

3. RESPONSIBILITY:

AR	Authorised Representative
HG	Head Grower

4. POLICY:

ILCO Farming will apply good agricultural practices for the drying of medicinal cannabis, as determined by the Department of Agriculture, Forestry and Fisheries (DAFF) for plants intended for human consumption, and Medicines and Related Substances Act 101 of 1965 to ensure all processes involved with the drying process are adhered to. Properly drying medicinal cannabis plants is paramount to prevent mould contamination from taking place.

5. PROCEDURE:

5.1 Before drying activity starts, the HG will ensure that the temperature and humidity data loggers are calibrated in place and set with the following parameters:

- Temperature: 18°C - 22°C.
- Humidity: 45% - 55%.

5.2 Cut 30-40 cm branches from each batch from the harvested plants.

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**
T

STANDARD OPERATING PROCEDURE

PROCEDURE FOR DRYING OPERATIONS

Page 2 of 3

SOP No:	3-CUL-006	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

- 5.3 Remove all unwanted leaves and then hang the branches from string or wire in the drying room, alternatively cut and hang the whole medicinal cannabis plant.
- 5.4 Leave a minimum space of 5cm between branches while drying.
- 5.5 Alternatively snip buds from branches and place them on medicinal cannabis drying racks.
- 5.6 Spread the trimmed material in one m² areas, in a thin layer on the drying racks, to ensure good air circulation around the plant material.
- 5.7 Drying of medicinal cannabis will take place in a low light designated drying room.
- 5.8 Drying will take between 12 and 18 days.
- 5.9 A daily track will be logged into the computer to record the status of the drying process.
- 5.10 Dried medicinal cannabis will be weighed per m².
- 5.11 The weight will be recorded into the batch processing documentation.

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter



ILCO
FARMING

Authorised
Copy

**ORIGINAL
DOCUMENT**

STANDARD OPERATING PROCEDURE

PROCEDURE FOR DRYING OPERATIONS

Page 3 of 3

SOP No:	3-CUL-006	Effective Date:	2020-06-11
Revision No:	2	Supersedes:	1
Revise By:	2022-06-11		

Attachment 6.1

SOP AMENDMENT HISTORY

VERSION NO:	SUPERSEDES:	EFFECTIVE DATE:	REASON FOR CHANGE:
1	N/A	2019-02-01	First Version
2	1	2020-06-11	Clarified procedure

COPY NO:		
Compiler:	Authorised by:	Approved by:
CCB – Nico Kriek	Coenraad Venter	Ilse Venter